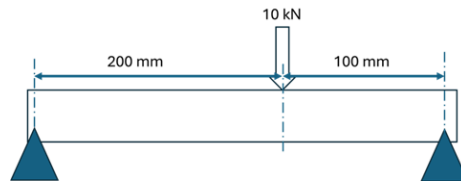


Tutorial 6: Shafts and Associated Elements

In-class activities

1. Find the maximum bending moment from the figure.



2. The key is fixed on the shaft with a key of dimension of 12 mm x 12 mm x 12 mm. Estimate the diameter of the shaft.
3. State the width and thickness of a square key for a shaft of diameter 80 mm.

Theory

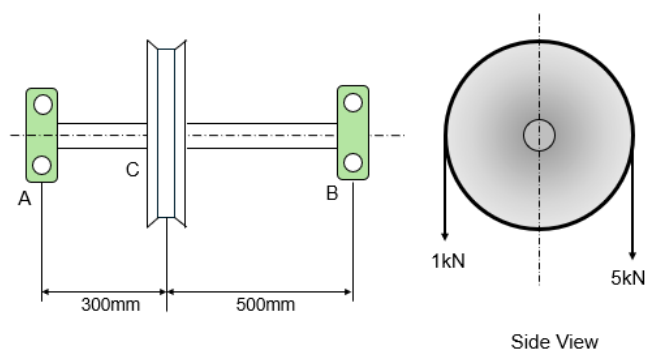
1. Define shaft's key and list two conditions that can affect the key's dimension.
2. Explain the main function and application of a coupling
3. List two types of pins used in shaft.
4. Explain the function and application of spline

Calculations

Question 1

A solid steel shaft with $S_y = 250\text{MPa}$, carries belt tensions vertically at pulley C, as shown in Figure below. Given diameter of pulley is 300mm, and a factor of safety of $n=1.5$, design the shaft according to the following failure criteria

- (i) Maximum Shear Stress Theory
- (ii) Maximum Energy of Distortion Theory



Example Solution

Given Data:

Steel shaft with $S_y = 250\text{MPa}$
 Pulley diameter, $d = 300\text{mm} = 0.300\text{m}$
 Pulley radius, $r = d/2 = 0.150\text{m}$
 Factor of safety, $n = 1.5$
 Belt tensions at pulley C: $T_1 = 5\text{ kN}$, $T_2 = 1\text{ kN}$

i - Design the shaft (determine diameter) using Maximum Shear Stress Theory of failure

Step 1: Analyse belt tensions at pulley. Find torque and forces acting on the shaft.

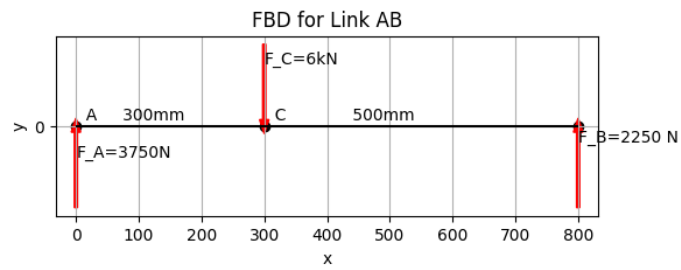
$$\begin{aligned} T_c &= r(F_1 - F_2) \\ &= 0.150(5000 - 1000) \\ &= 600\text{Nm} \end{aligned}$$

Total load at C, $F_C = F_1 + F_2 = 5\text{ kN} + 1\text{ kN} = 6\text{ kN}$

Step 2: Calculate the reactions at the supports A and B

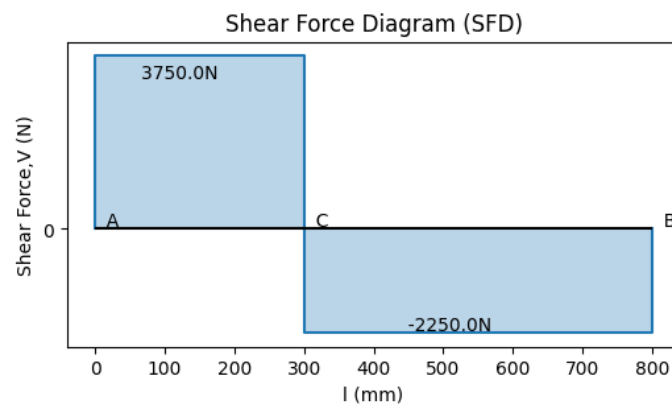
Draw the free body diagram of the shaft:

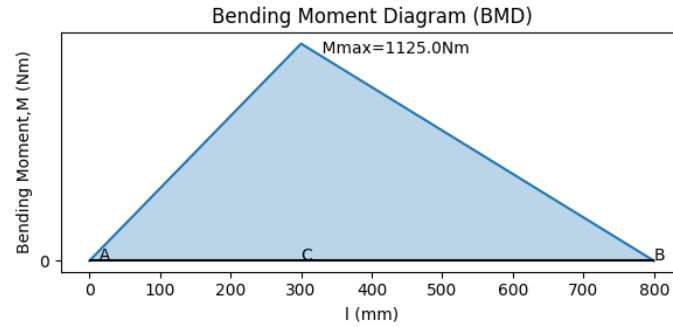
Taking moments about A to find reaction at B:



$$\begin{aligned}
 + \circlearrowleft \sum M_A &= 0 \\
 -F_C(0.3) + F_B(0.8) &= 0 \\
 -(6000)(0.3) + F_B(0.8) &= 0 \\
 F_B &= 2250N \\
 + \uparrow \sum F_y &= 0 \\
 F_A + F_B - F_C &= 0 \\
 F_A + 2250 - 6000 &= 0 \\
 F_A &= 3750N
 \end{aligned}$$

Step 3: Draw shear force and bending moment diagrams to find maximum bending moment.





Maximum bending moment, $M_{max} = 1125 \text{ Nm}$

Step 4: Calculate the diameter of the shaft using Maximum Shear Stress Theory of failure.

i - Design the shaft (determine diameter) using Maximum Shear Stress Theory of failure From Maximum Shear Stress Theory of failure:

$$\begin{aligned}\frac{S_y}{n} &= \frac{32}{\pi D^3} (M^2 + T^2)^{\frac{1}{2}} \\ D^3 &= \frac{32n}{\pi S_y} (M^2 + T^2)^{\frac{1}{2}} \\ D^3 &= \frac{32(1.5)}{\pi(250 \times 10^6)} [(1125)^2 + (600)^2]^{\frac{1}{2}} \\ &= 7.7922 \times 10^{-5} m^3 \\ D &= 0.04271 m = 42.7 mm\end{aligned}$$

The minimum diameter of the shaft is 43mm.

ii - Design the shaft (determine diameter) using Maximum Energy of Distortion Theory of failure From Maximum Energy of Distortion Theory of failure:

$$\begin{aligned}\frac{S_y}{n} &= \frac{16}{\pi D^3} \left[(M^2 + \frac{3}{4} T^2) \right]^{\frac{1}{2}} \\ D^3 &= \frac{16n}{\pi S_y} \left[(M^2 + \frac{3}{4} T^2) \right]^{\frac{1}{2}} \\ &= \frac{16(1.5)}{\pi(250 \times 10^6)} \left[(1125)^2 + \frac{3}{4} (600)^2 \right]^{\frac{1}{2}} \\ &= 7.573 \times 10^{-5} m^3 \\ D &= 0.04231 m = 42.3 mm\end{aligned}$$

The minimum diameter of the shaft is 43mm.

Question 2

A shaft shown in the Figure Q2 moves at 16rpm to transmit 1kW power from pulley B to Pulley A. Design the shaft according to Maximum shear stress theory of failure $S_y=550\text{MPa}$ and $n=2$.

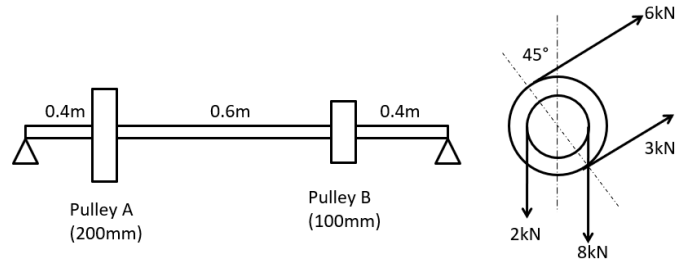


Figure Q2

Question 3

Figure Q3 shows a shaft loaded with pulleys C and D, supported by bearings A and B. The shaft transmits 70 kW of power at 1500 rpm to both pulleys. The diameters of pulleys C and D are 150 mm and 100 mm, respectively. Meanwhile, the belt tension ratios for pulley C and D are 2:1 and 3:2, respectively. The high-strength steel with a yield stress of $S_y=345$ MPa is used for the shaft, with a safety factor of 2.5

- (i) Determine the belt tension on pulley C and D.
- (ii) Determine the diameter of the shaft using maximum shear stress theory of failure.
- (iii) If the material of the shaft changes to 600 MPa of the yield stress, suggest the new diameter of the shaft.

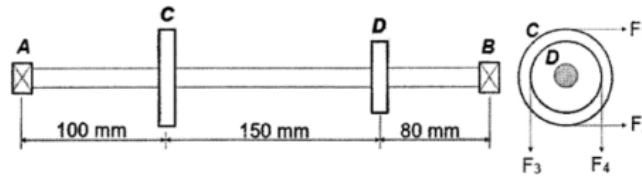


Figure Q3

Question 4

The layout of a shaft carrying two pulleys A and C and supported by two bearings B and D is shown Figure Q4. The shaft transmits power at 360 rpm from pulley A to pulley C. The diameter of pulley A and C are 200 mm and 300 mm respectively. The belt tensions act horizontally at pulley C and the ratio of belt tension on the tight side to the slack side is 2.5:1. The shaft is made of plain carbon steel with a yield strength of 280MPa and the factor of safety is 2.

- (i) Determine the belt tensions on pulley C.
- (ii) Determine the diameter of the shaft using maximum shear stress theory of failure.
- (iii) What is the new diameter of the shaft if the factor of safety is increased to 3.

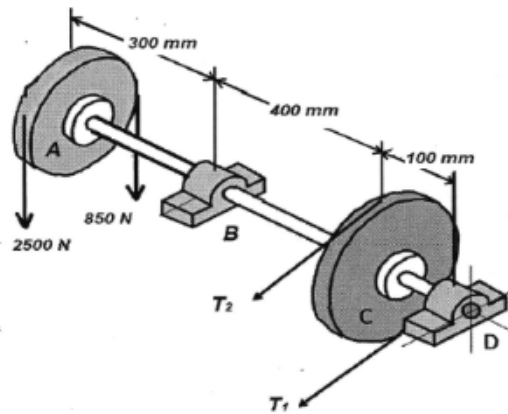


Figure Q4

Answer

Q2 $d = 45\text{mm}$

Q3

(i) $T_1 = 11883.46\text{ N}$, $T_2 = 5941.73\text{ N}$, $T_3 = 26737.8\text{ N}$, $T_4 = 17825.2\text{ N}$

(ii) $d = 59\text{mm}$

(iii) $d = 49\text{mm}$

Q4

(i) $T_1 = 1834.8\text{ N}$, $T_2 = 733.9\text{ N}$

(ii) $d = 42\text{mm}$

(iii) $d = 48\text{mm}$